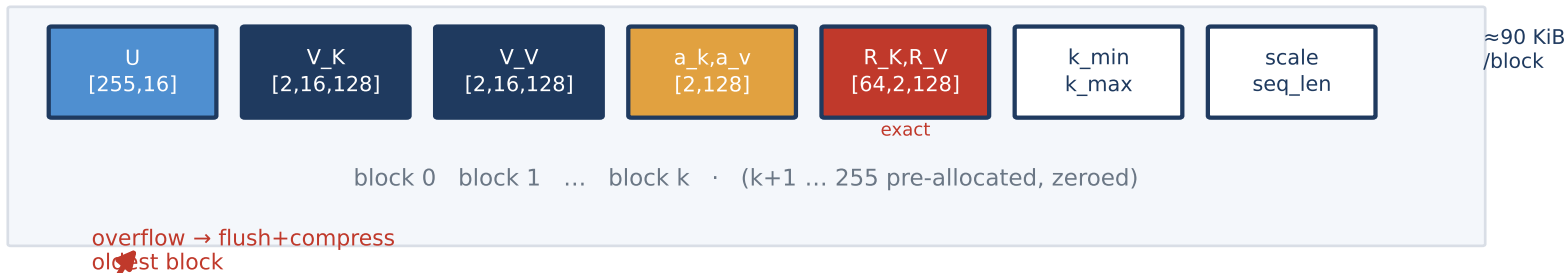


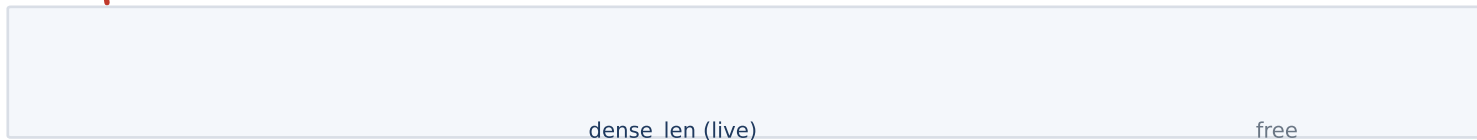
Per-Session KV Store — memory layout (one layer of 28)

Bounded dense recency window + fixed compressed pool — the size that does NOT grow with context

Compressed pool (fixed $M = 256$ blocks · $\leq 65\,536$ tokens)



Dense recency window (sliding, capacity $W+B = 768$ tokens, uncompressed fp16)



Footprint per layer $\approx$ dense $768 \cdot H_{kv} \cdot d \cdot 2 \cdot 2 \cdot B$ + used_blocks · 90 KiB.

Dense window is constant; the compressed term grows at $\approx 1/2.85$ the dense-KV rate, and the pool is capped → the bounded memory slope.